Mohammed Bilal N S

Full Stack Developer

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Profile Summary

Full-Stack Developer with experience building secure, scalable web applications and backend systems. Skilled in designing REST APIs, authentication and authorization, relational databases, and maintainable software using clean software architecture and modern development practices.

Technical Skills

Languages: JavaScript, TypeScript, Python, Go
Frontend: React.js, Next.js, TanStack Start, Redux, TanStack Query
Backend: Node.js, Express.js, Fastify, NestJS, RESTful APIs, WebSockets, JWT Authentication, REST API Design
Databases: MongoDB, PostgreSQL, Redis
Cloud & DevOps: Docker, AWS (EC2, S3), CI/CD(GitHub Actions), Nginx
Dev Tools: Git, GitHub, Cursor, OpenAI Codex CLI, Antigravity CLI
Integrations: Stripe, PayPal, Razorpay, Cloudinary

Projects

- **Upstride (Mentorship & Expert Consultation Platform)** Live Link - GitHub

  A mentorship platform with mentor discovery, scheduling, payments, real-time video/chat, and article publishing.

  Tech stack: React, TypeScript, Node.js, Express, MongoDB, Redis, Socket.IO, WebRTC, Stripe, Docker, AWS

  - Backend with **Clean Architecture**, repository pattern, JWT authentication, RBAC, Redis caching, and BullMQ background jobs for scalability and maintainability
  - Mentor workflow with recommendations, search/filtering, availability scheduling, booking/rescheduling, Stripe and wallet payments, invoicing, and automated reminders
  - Real-time **WebRTC** video, Socket.IO messaging, collaborative whiteboard, article publishing with recommendations and moderation/appeal workflows, admin modules for verification and analytics; deployed with Docker on **AWS EC2**

- **TechHive (PC Accessories E-Commerce Platform)** Live Link - GitHub

  A full-featured e-commerce platform with payments, inventory, promotions, wallets, and order management.

  Tech stack: Node.js, Express.js, MongoDB, EJS, Tailwind CSS, Razorpay

  - **MVC backend** with product catalog, inventory tracking, cart, wishlist, and end-to-end order processing
  - Product discovery with filtering, sorting, categories, and reviews, plus coupons, offers, discounts, and wallet for refunds/purchases
  - **Razorpay** payments, authentication, profile management, and order tracking; deployed on **AWS EC2**

- **Lumina (Content Publishing & Article Discovery Platform)** Live Link - GitHub

  A content publishing platform with article creation, personalized recommendations, discovery, and media management.

  Tech stack: NestJS, PostgreSQL, Drizzle ORM, TanStack Start

  - **NestJS** + **TanStack Start** with JWT authentication, publishing workflows, and media management
  - Personalized recommendations, search, filtering, and interest-based content discovery
  - **PostgreSQL** + **Drizzle ORM** data model with secure **AWS S3** media uploads

Mini Projects

CineMind – AI-Powered Movie Recommendation Platform Live Link - GitHub

Built an AI-powered movie recommendation platform leveraging LangChain and TMDB to generate personalized recommendations with contextual explanations and rich movie metadata.

Tech: Fastify, TypeScript, LangChain, Zod, TMDB API

ShrinkLink – URL Shortener & Analytics Platform Live Link - GitHub

Developed a URL shortening platform with real-time visit tracking, click analytics, and redirect management, enabling users to monitor link performance and engagement.

Tech: Go, React, MongoDB

Experience

- **Freelance Full Stack Developer** Apr 2025 – Sep 2026
  - Designed, built, and deployed responsive full-stack web applications using React, Node.js, Express, and MongoDB with secure REST APIs and JWT authentication.
  - Collaborated directly with clients to gather requirements, ship production-ready features on AWS and custom VPS, and ensure performance, maintainability, and on-time delivery.
- **Freelance Mathematics Tutor & LaTeX Typesetter** Nov 2023 – Mar 2025
  - Prepared publication-quality academic documents, dissertations, and mathematical content using LaTeX while tutoring undergraduate mathematics with a focus on accuracy and clear presentation.

Education

- **Professional Training – Full-Stack Development (Brototype)** October 2024 – June 2026

  Completed an intensive project-based bootcamp focused on React, Node.js, TypeScript, REST APIs, PostgreSQL, MongoDB, Docker, Git, AWS, and modern software development practices.

- **Master of Science (M.Sc.) in Mathematics** 2021 – 2023

  TKM College of Arts and Science, Kollam

  CGPA: **7.9/10**

- **Bachelor of Science (B.Sc.) in Mathematics** 2018 – 2021

  TKM College of Arts and Science, Kollam

  CGPA: **8.3/10**

Languages

English, Malayalam